

Evaluating Saliency Methods in Medical Vision-Language Models

Mikhail Belov

*Department of Advanced Computing Sciences
Faculty of Science and Engineering
Maastricht University
Maastricht, The Netherlands*

Abstract—Visual saliency methods have been developed to explain model decision making, and in a field such as medicine, where decisions can have serious consequences, explainability is crucial for trust and adoption. While saliency methods have been benchmarked on Vision Transformer (ViT) classifiers in medical tasks, they have not been systematically evaluated on medical vision-language models (VLMs). This thesis adapts three explainability methods, Attention Rollout, GMAR, and Grad-CAM, to VLM architectures and evaluates them using a teacher-forced deletion experiment on the ROCov2 radiology dataset with two medical VLMs: MedGemma and LLaVA-Med. The same framework is then used to quantify the extent to which generated medical captions are visually grounded. The deletion results show that, for both models, the saliency methods produce larger probability drops than random masking, suggesting that the methods effectively identify image regions relevant to generated words. Attention Rollout and GMAR perform almost identically across both models, while Grad-CAM generally underperforms the rollout methods. An ablation on MS-COCO, a general captioning dataset, and on the corresponding base models (Gemma 3 and LLaVA-1.5) shows that AOPC is strongly affected by dataset domain, with most models producing larger probability drops on the general captioning dataset.

Index Terms—Attention Rollout, Explainable AI (XAI), Vision Transformer, Vision-Language Models, Grad-CAM

I. INTRODUCTION

A key challenge in the widespread clinical adoption of AI models is that they are largely black-box. It is difficult to trust a model with a diagnosis if there is no way to understand why it produced a given output. The field of explainable AI (XAI) has tackled model explainability with saliency methods which are post-hoc algorithms applied to image-processing model architectures to produce heatmaps over the input. These heatmaps are designed to highlight the regions of the image that the model relied on when assigning a particular class, and Figure 1 illustrates an example from the experiments in this thesis. Saliency methods have been developed for many model architectures, with the most recent literature targeting Vision Transformers [1] (ViTs). A ViT adapts the transformer architecture to images by splitting the input into a grid of patches, embedding each patch as a token, and processing the resulting sequence through self-attention layers, with a prepended CLS token whose final hidden state is read out by

This thesis was prepared in partial fulfilment of the requirements for the Degree of Bachelor of Science in Data Science and Artificial Intelligence, Maastricht University. Supervisor(s): Dr. Michel Dumontier, Dr. Chang Sun

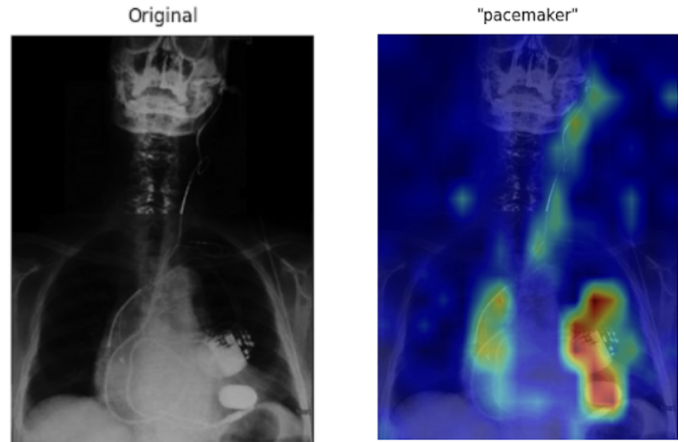


Fig. 1. Example saliency map produced for a single generated token by MedGemma. *Left*: input image. *Right*: attention-rollout heatmap for the generated token "pacemaker", overlaid on the input. The heatmap localises onto the pacemaker.

a classification head. The three saliency methods evaluated in this thesis, Grad-CAM [2], Attention Rollout [3] and Gradient-driven Multi-head Attention Rollout (GMAR) [4], were all adapted and validated for this setting.

However, when it comes to industry adoption of AI models in medicine, the next frontier is Vision-Language Models (VLMs) rather than classifiers. A VLM is, at the architectural level, a Large Language Model (LLM) coupled with a ViT acting as the vision encoder, enabling multimodal input of image and text into a language model. Several VLMs have been trained for clinical tasks. However, the saliency methods described above were designed and validated for classifiers, and their behaviour on medical VLMs has not been systematically evaluated. This study aims to bridge this gap by evaluating how Attention Rollout, GMAR and Grad-CAM perform when applied to two medical VLMs, MedGemma [5] and LLaVA-Med [8] on the ROCov2 [7] radiology dataset. A deletion test is used to evaluate how effectively these methods identify the important regions in the image, and an ablation test was carried out to study the effect of modality on visual grounding. This thesis aims to answer the following questions:

- 1) How effective are different explainability methods such

as GMAR, Grad-CAM, and Attention Rollout at identifying important image regions in medical VLMs?

- 2) To what extent do medical VLMs ground themselves in visual input versus language priors when generating medical captions?

A. Prior Work

Saliency methods have been benchmarked on classifiers using both general, and most importantly for this study, medical datasets. Saporta *et al.* [18] benchmark seven saliency methods on chest-X-ray pathology classifiers and find that the best methods substantially underperform expert localisation. Arun *et al.* [19] report similar findings on classifiers spanning CT, MRI and X-ray. DeGrave *et al.* [20] use saliency methods to discover that COVID-19 radiograph classifiers were attending to image metadata rather than pulmonary signal, establishing saliency as a shortcut-detection tool in radiology. Closest to the methods used in this thesis, Barekatin and Glocker [21] quantitatively compare Gradient Attention Rollout and Grad-CAM on several ViT backbones (ViT, DeiT, DINO, Swin) on blood cell and breast ultrasound classification, reporting that Grad-CAM produces more localised and class-discriminative attribution than Gradient Attention Rollout on those tasks. These studies all target classifiers and span a limited set of medical datasets. The equivalent question for generative medical VLMs has not yet received the same systematic treatment.

II. METHODS

A. Vision-Language Models

Both models share the same architectural pattern: a ViT serves as a visual encoder, producing per-patch embeddings that are projected into the hidden space of an autoregressive LLM. The image is processed by splitting it into fixed-size patches, projecting each into a 1D embedding via the ViT’s multi-head attention layers, and inserting these patch embeddings as tokens into the LLM’s input sequence.

MedGemma is a clinical VLM developed by Google, built on the Gemma 3 [6] multimodal language model backbone. Its language model is a 34-layer decoder, and its vision encoder is MedSigLIP, a SigLIP-2 [10] variant pre-trained on medical images. LLaVA-Med is an open-source VLM derived from LLaVA-1.5 [13] and adapted for biomedical visual question answering. The version used in this study (LLaVA-Med v1.5, Mistral-7B variant) couples CLIP ViT-L/14 [11] visual encoder with a Mistral-7B [9] autoregressive language model with a 32-layer decoder.

B. Explainability Methods

In the ViT formulations of the saliency methods explored in this study, the methods target a single class logit produced by the classifier. When porting these methods to generative VLMs, three adaptations are required:

- 1) The target is a per-token next-token logit at a specific position p in the generated sequence, rather than a per-class output logit. A separate saliency map is produced for each generated token.

- 2) The attention matrices and intermediate activations used by the methods come from the LLM’s causal decoder rather than from a bidirectional ViT encoder.
- 3) The saliency for the token at position p is read as the row of the resulting relevance matrix at position p , restricted to the sequence positions occupied by image-patch tokens, and reshaped to the spatial grid implied by the patch count (24×24 for LLaVA-Med, 16×16 for MedGemma).

a) **Attention Rollout and GMAR:** Both methods compose per-layer attention matrices across a uniformly spaced subset of decoder layers $\mathcal{L}$ (every fourth for LLaVA-Med, every sixth for MedGemma) to mitigate attention collapse. For layer l with H total attention heads, the combined attention matrix $A^{(l)}$ is a weighted sum:

$$A^{(l)} = \sum_{h=1}^H w_h^{(l)} A_h^{(l)}$$

Here, $A_h^{(l)}$ represents the attention matrix for an individual head h , and $w_h^{(l)}$ denotes its assigned weight. In Attention Rollout, heads are weighted uniformly ($w_h^{(l)} = 1/H$). GMAR derives $w_h^{(l)}$ by computing the gradient of the next-token logit $y_t^{(p)}$ (the prediction for target id t at position p) with respect to each head ($G_h^{(l)}$), calculating its L1 or L2 norm, retaining only non-negative values, and L1-normalising across heads. The rollout step comes from blending $A^{(l)}$ with the identity matrix and row-normalising. The cumulative rollout matrix A_{rollout} is the matrix product of these adjusted layers over $\mathcal{L}$. Saliency for token p is extracted from $A_{\text{rollout}}[p, \cdot]$ at the image-patch columns.

b) **Grad-CAM:** Grad-CAM targets the activation matrix $A \in \mathbb{R}^{N \times D}$ hooked at the multimodal projector output. For the generated token at position p with target id t , importance for each patch position $i \in \{1, \dots, N\}$ is computed as:

$$L_i = \text{ReLU} \left(\sum_{k=1}^D \frac{\partial y_t^{(p)}}{\partial A_{i,k}} \cdot A_{i,k} \right)$$

Here, N defines the total number of visual patch embeddings, D represents the embedding dimension size, and $A_{i,k}$ is the specific activation value for patch i at dimension k with ReLU isolating positive contributions to the prediction.

The three methods together cover pure attention, pure gradient, and hybrid approaches, providing a broad overview of how different approaches isolate patch importance in this setting.

C. Evaluation Metric: Perturbation via Teacher Forcing

To quantitatively evaluate the faithfulness of the generated heatmaps, this thesis uses the Area Over the Perturbation Curve (AOPC) [16]. AOPC measures the decline in a model’s prediction confidence as the most salient visual features, identified by the respective explainability method, are progressively removed.

As VLMs are autoregressive, the probability of predicting a token depends on its position in the sequence. Unbounded generation allows the model to continuously alter its output sequence, making token-level attribution unstable. To isolate the model’s internal state, teacher forcing [17] is applied. Given an original image I and a generated sequence of tokens $S = \{t_1, t_2, \dots, t_n\}$, the preceding tokens $S_{<i} = \{t_1, t_2, \dots, t_{i-1}\}$ are provided to the model as a fixed language prior.

Perturbations are applied via a zero¹ masking strategy. For a given token t_i and a masking ratio r , the top r fraction of the most salient regions is replaced to create a perturbed image, denoted as $\tilde{I}_{i,r}$. The probability drop, $\Delta P(t_i, r)$, is then calculated as the difference between the initial probability of generating token t_i and the probability of generating t_i given the perturbed image:

$$\Delta P(t_i, r) = P(t_i | I, S_{<i}) - P(t_i | \tilde{I}_{i,r}, S_{<i})$$

The final AOPC is calculated as the average probability drop across a predefined sequence of masking ratios.

III. EXPERIMENTS

Two experiments were conducted: the main experiment evaluating the explainability methods for the two medical VLMs, and an ablation experiment which involved the two medical VLMs as well as their base models, LLaVA-1.5 and Gemma 3. In the ablation experiment, two datasets were used: ROCoV2 and MS-COCO [12], a general captioning dataset. Both experiments were conducted on an NVIDIA RTX PRO 6000 GPU with 96GB of VRAM. The sample included 500 images from the ROCoV2 dataset and 100 from MS-COCO.

A. Evaluating Explainability Methods in Medical VLMs

The medical VLMs received an image I from the ROCoV2 dataset alongside the prompt: “Generate a single-sentence medical caption for this radiology image”. To evaluate the faithfulness of the generated heatmaps for every token t_i across the output sequence, the AOPC metric described in Section II-C was applied. Perturbations were applied using masking ratios $r \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$. Additionally, a random map is generated and perturbed in the same manner, this is the random baseline the saliency methods are compared to.

B. Ablation Experiment

To disentangle the contribution of the medical fine-tuning and medical language to the probability drops, the same perturbation procedure as in III-A was run on a 2×2 ablation: each medical VLM and its non-medical base (MedGemma vs. Gemma 3; LLaVA-Med vs. LLaVA-1.5) were evaluated on both ROCoV2 and MS-COCO. In the MS-COCO condition, the prompt was: “Generate a caption for this image.”

¹Zero masking means replacing the selected image patches with zero-valued pixels before re-running the model.

C. Data Processing

a) *Word Reconstruction From Tokens*: Both MedGemma and LLaVA-Med employ subword tokenisation, which splits words into multiple tokens, for example, “tomography” may be tokenised as `tom`, `ography`. In a subword sequence like `tom` and `ography`, the true probability drop occurs at `tom`, since `ography` is predicted with a probability of 1. To obtain more truthful saliency scores, these sub-tokens are merged into whole words whose boundaries are detected via the tokenizer’s word-initial markers. Punctuation tokens are removed before statistical analysis, as punctuation is a language artefact.

b) *Clinical Named-Entity Recognition*: To assign each reconstructed word a semantically meaningful label, a named entity recognition (NER) model is applied. This thesis uses Clinical-AI-Apollo/Medical-NER [14], a DeBERTa-v3 [15]-based token classifier fine-tuned on biomedical and clinical corpora to predict medical entity types. Each merged word is then aligned to an entity group. Words that do not overlap any entity (function words, stopwords, residual tokens) are tagged as `OTHER`. This produces a single categorical label per word, which is subsequently used to stratify the faithfulness analysis by semantic class and to test whether saliency methods preferentially attribute probability mass to clinically relevant words.

IV. RESULTS

A. AOPC and Distribution

Table I shows that for MedGemma, the four informed methods clear the random baseline by ≈ 2 times. The attention rollout methods are clustered together and all three outperform Grad-CAM. Figure 3 shows that the mean drop has a positive relationship with the mask ratio r and that the gap between Grad-CAM and the rollout methods is smallest at the $[0.1-0.2]$ ratio, with the gap widening as r increases. For LLaVA-Med, the AOPC scores are lower across the board, and the informed methods are much closer to the random baseline based on AOPC. Looking at the per r drops, at 0.1 there is a much larger difference between the random baseline and the informed methods; for $r \geq 0.4$, Grad-CAM performs worse than random. A similar relationship between Grad-CAM and the rollout methods can be observed and the rollout methods are also clustered together. Figure 4 shows that the distribution of probability drops across all informed methods does not follow a normal distribution.

B. Base-Model Ablation

Table II shows the Attention AOPC for all combinations. The Gemma-family models and LLaVA-1.5 show larger drops on MS-COCO than on ROCoV2, while LLaVA-Med is an exception in these runs, with a very small COCO AOPC. As discussed in Section V, this value partly reflects poor generation quality under the MS-COCO prompt.

C. Statistical Significance

To test whether the five methods produce different per-word drops, this thesis uses the Friedman test, a non-parametric

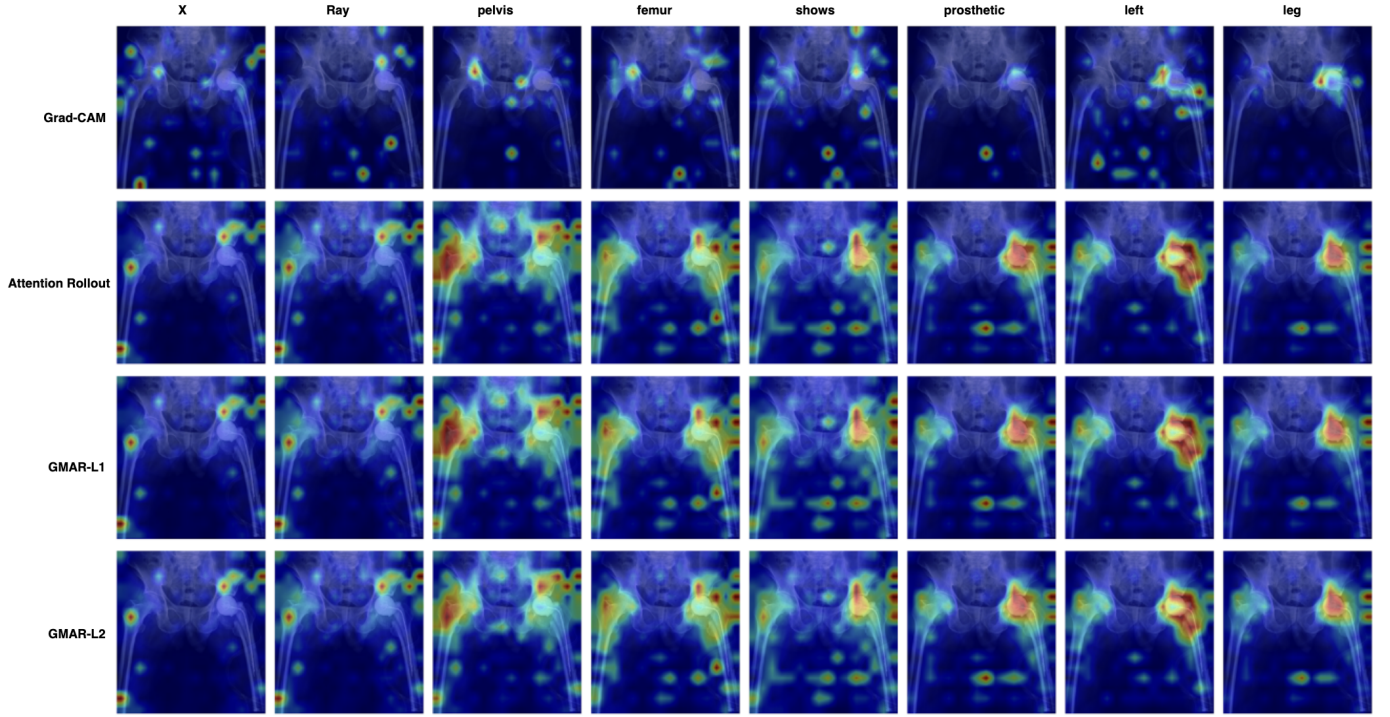


Fig. 2. Example of the saliency maps generated by each method taken from a random ROCov2 sample with stopwords removed.

TABLE I
MEAN PER-WORD PROBABILITY DROP $\overline{\Delta p}$ BY XAI METHOD AND MASK RATIO FOR MEDGEMMA (TOP) AND LLaVA-MED (BOTTOM), FILTERED FOR OTHER. AOPC IS THE MEAN ACROSS THE FIVE MASK RATIOS.

MedGemma						
Method	0.1	0.2	0.3	0.4	0.5	AOPC
Attention	0.0941	0.1142	0.1280	0.1414	0.1562	0.1268
GMAR-L1	0.0938	0.1143	0.1281	0.1418	0.1567	0.1269
GMAR-L2	0.0936	0.1136	0.1275	0.1409	0.1560	0.1263
Grad-CAM	0.0865	0.1068	0.1190	0.1294	0.1401	0.1164
Random	0.0161	0.0348	0.0572	0.0788	0.1031	0.0580

LLaVA-Med						
Method	0.1	0.2	0.3	0.4	0.5	AOPC
Attention	0.0205	0.0362	0.0508	0.0648	0.0750	0.0494
GMAR-L1	0.0201	0.0358	0.0505	0.0648	0.0753	0.0493
GMAR-L2	0.0203	0.0359	0.0507	0.0648	0.0755	0.0494
Grad-CAM	0.0195	0.0337	0.0419	0.0493	0.0579	0.0404
Random	0.0121	0.0253	0.0397	0.0530	0.0703	0.0401

TABLE II
ATTENTION AOPC FOR MEDICAL VLMS AND THEIR BASE MODELS ON ROCov2 AND COCO, COMPUTED ON THE FIRST 100 SAMPLES FOR EACH DATASET. ALL VALUES USE WORD-LEVEL DROPS. FOR ROCov2, WORDS TAGGED OTHER BY MEDICAL NER ARE REMOVED; FOR COCO, PURE PUNCTUATION AND STOPWORDS ARE REMOVED INSTEAD.

Dataset	Attention AOPC			
	MedGemma	Gemma 3	LLaVA-Med	LLaVA-1.5
ROCov2	0.1105	0.1757	0.0515	0.0674
COCO	0.2802	0.3058	0.0120	0.1490

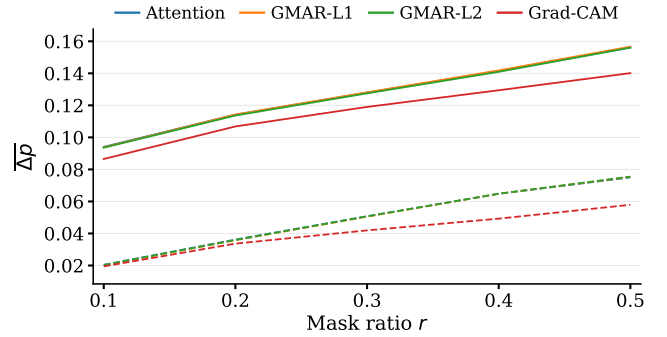


Fig. 3. Mean per-word probability drop $\overline{\Delta p}$ versus mask ratio r for every informed method on both models (filtered: NER class OTHER removed). Solid lines: MedGemma. Dashed lines: LLaVA-Med.

repeated-measures test that, within each (mask ratio, NER class) stratum, ranks the methods per word and asks whether the rank sums differ more than expected by chance. Because the five methods are evaluated on the same words, the test is paired and makes no normality assumption. With 5 mask ratios and one stratum per NER class, this yields 50 strata on MedGemma (10 classes) and 55 on LLaVA-Med (11 classes); after Benjamini–Hochberg correction at $q < 0.05$, the Friedman test rejects equality of methods in 41/50 strata on MedGemma and 27/55 on LLaVA-Med. As the mask ratio grows the informed methods converge towards the random baseline. The effect is strongest on LLaVA-Med, where at $r = 0.5$ the rollout family (0.075) is within 7% of the random

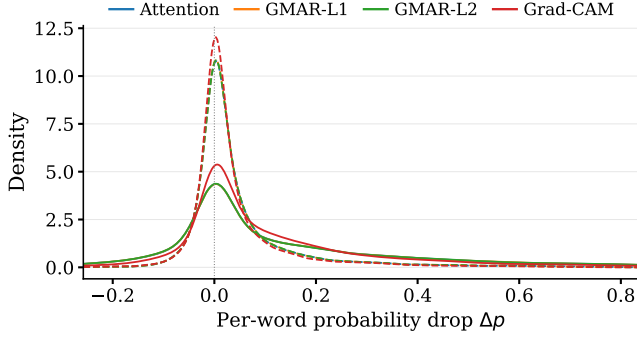


Fig. 4. Kernel-density estimate of the per-word probability drop Δp for each informed method on both models, pooled over all mask ratios (filtered: NER class OTHER removed). Solid lines: MedGemma. Dashed lines: LLaVA-Med. The x-axis is truncated to the 1st–99th percentile range of the pooled drops.

baseline (0.070), whereas on MedGemma the same gap is still 0.156 vs. 0.103 (51%).

D. Clinically Stratified Attribution

On MedGemma the largest drops occur for SEVERITY (≈ 0.28) and THERAPEUTIC_PROCEDURE (≈ 0.23), followed by DETAILED_DESCRIPTION (≈ 0.21). Attention, GMAR-L1, and GMAR-L2 are nearly identical within each class, while Grad-CAM is lower on classes such as THERAPEUTIC_PROCEDURE. On LLaVA-Med the same clinically stratified drops are much smaller: BIOLOGICAL_STRUCTURE is the largest class for the rollout family (≈ 0.071), while SIGN_SYMPTOM is near zero and SEVERITY is slightly negative.

TABLE III
AOPC BY NER ENTITY CLASS FOR EACH INFORMED METHOD.

MedGemma				
Entity class	Attention	GMAR-L1	GMAR-L2	Grad-CAM
Bio. Struct.	0.131	0.131	0.131	0.127
Diag. Proc.	0.042	0.043	0.042	0.043
Therap. Proc.	0.233	0.232	0.231	0.101
Det. Desc.	0.207	0.207	0.205	0.170
Dis./Disorder	0.086	0.086	0.085	0.081
Sign/Sympt.	0.104	0.104	0.105	0.099
Severity	0.279	0.278	0.278	0.235
Lab Value	0.100	0.097	0.097	0.091
LLaVA-Med				
Entity class	Attention	GMAR-L1	GMAR-L2	Grad-CAM
Bio. Struct.	0.071	0.071	0.071	0.059
Diag. Proc.	0.034	0.034	0.034	0.027
Therap. Proc.	0.065	0.065	0.064	0.044
Det. Desc.	0.045	0.046	0.046	0.036
Dis./Disorder	0.032	0.032	0.032	0.030
Sign/Sympt.	0.009	0.009	0.009	0.010
Severity	-0.003	-0.003	-0.003	-0.003
Lab Value	0.020	0.020	0.021	0.028

A. Evaluating Explainability Methods

The results show that Attention Rollout, GMAR, and Grad-CAM identify image regions that are more relevant than random masking under the AOPC metric. However, AOPC alone is not a complete indicator of explanation quality: as the mask ratio increases, different heatmaps overlap because a large fraction of the image is removed. The lower mask ratios are therefore more informative for method comparison. In that regime, the probability drops from the informed methods are roughly twice as large as the random baseline on LLaVA-Med and almost six times as large on MedGemma. Across both models, the two rollout variants are nearly indistinguishable, while Grad-CAM generally underperforms them.

This ordering differs from deletion results reported for ViT classifiers, where GMAR can outperform both Attention Rollout and Grad-CAM, and Grad-CAM can outperform plain Attention Rollout [4]. Prior deletion studies report relatively small differences between saliency methods so we should not expect large performance disparities. These experiments evaluate over masking ratios spanning $r \in [0, 1]$ [2, 4]. Here, a narrower masking range is used as well as different datasets, teacher-forced word probabilities rather than class logits, and VLMs rather than standalone ViT classifiers. Direct numerical comparison to the ViT literature is therefore not appropriate.

a) Similarity Between Attention Rollout and GMAR:

Attention Rollout can be viewed as the unweighted case of GMAR: it averages all attention heads equally, whereas GMAR first weights each head by a gradient-based importance score. In the ViT literature, this distinction can lead to a clear performance difference, with GMAR achieving stronger deletion scores than plain Attention Rollout. This is plausible in ViTs because the attribution path is relatively direct: gradients are taken from the class score back through a bidirectional image encoder, where patch tokens are the primary representational units and different heads often capture distinct spatial relations. As a result, the gradient norms assigned to different heads can remain meaningfully non-uniform, so GMAR does not collapse to the simple head average. In the VLM setting considered here, the situation is different. The target is a generated word probability, and the attribution signal must propagate backward through a deep causal language decoder before reaching the image-token positions. This decoder contains 32 layers in LLaVA-Med and 34 layers in MedGemma. During this propagation, the gradient signal is repeatedly mixed across many preceding text and image positions by causal self-attention. The result is that the per-head L1 and L2 gradient norms become nearly identical. After normalisation, the GMAR head weights w_h therefore approach the uniform value $1/H$, where H is the number of heads. In this regime, GMAR’s weighted head sum becomes almost equivalent to the unweighted head mean used by plain Attention Rollout, explaining why the two methods produce nearly identical AOPC values in this study.

b) *Grad-CAM*: The same gradient-flattening may also contribute to Grad-CAM’s underperformance. Grad-CAM relies entirely on the gradient of the target logit with respect to a hooked feature map. In this setup, that gradient must traverse the causal decoder, where each layer aggregates information from earlier positions. This can reduce the localisation of the per-position attribution signal. The effect is compounded by the hook location: the multimodal projector output is already downstream of CLIP or MedSigLIP image encoding, where image patch information has been mixed by bidirectional attention layers. As a result, gradients at the projector do not necessarily map cleanly back to individual image regions.

Different results could have been obtained by selecting another Grad-CAM hook target or another subset of decoder layers for the rollout methods. The observations above therefore apply to this experimental setup and should not be treated as a general statement regarding all medical VLMs. The following subsections discuss additional design choices that affect how the AOPC scores should be interpreted.

B. Teacher Forcing

Using teacher forcing means that this experiment does not measure free-form reliance on the image. Rather, it measures the marginal contribution of image content to next-token prediction, conditional on the model having perfect prior context. VLMs are large language models with an attached visual encoder, and giving a medically fine-tuned model a prior such as “... that shows acute” and expecting it to generate anything other than “appendicitis” would not be a meaningful test of visual grounding, regardless of the image content. When the visual signal is weakened by perturbation, both VLMs default to the language priors that produced the reference caption in the first place, and the AOPC drop is therefore bounded above by the margin of contribution the image made over and above those priors. This bound may be tighter on LLaVA-Med than on MedGemma because LLaVA-Med’s Mistral-7B backbone was trained as a text-only language model, while MedGemma’s Gemma 3 backbone was trained multimodally from the start. MedGemma is therefore expected to rely more directly on visual tokens, and the substantially larger AOPC values observed for it are consistent with this architectural distinction. Teacher forcing was necessary to enable per-token evaluation: a free-form decoding protocol would allow the model to deviate from the reference caption at each step and would prevent a direct probability comparison between perturbed and unperturbed inputs. A complementary protocol that generates captions with and without the image and compares the resulting sequences would measure overall image dependence rather than the marginal contribution measured here, and is left to future work.

C. Inter-Model Comparisons and Generalisation

A direct comparison between results generated by LLaVA-Med and MedGemma is not meaningful as these models are very different despite sharing the same overall VLM architecture. MedGemma is a newer model (2025) compared

to LLaVA-Med (2023), and it has a medically fine-tuned ViT encoder. LLaVA-Med is the product of an experiment showing that LLaVA-1.5 could be adapted to the biomedical domain within a small training budget [8]. On this benchmark, the two models should not be treated as equally representative data points. MedGemma is a better representative of the state-of-the-art but even then, no generalisation should be made to the whole field of medical VLMs on the basis of the results obtained.

D. Modality

The low AOPC scores observed on ROCov2 are jointly attributable to the teacher-forcing protocol and to the content-heavy nature of medical captioning. Medical captions are dominated by methodology, anatomy and templatic phrasing, each of which is largely predictable from a perfect language prior alone, so the marginal contribution of image content to next-token prediction is inherently smaller for medical captions than for general-domain ones. This effect is directly visible in Table II: three of the four models produced AOPC values roughly two to three times higher on MS-COCO than on ROCov2. LLaVA-Med is the exception. An inspection of the generated MS-COCO outputs shows that many of its captions were not clean image descriptions: 23 of the 100 sampled outputs contain instruction-template text such as “artificial intelligence assistant”, and some outputs degenerate into repeated fragments such as repeated “toothpaste box” or “wooden box” phrases. The unusually low LLaVA-Med COCO AOPC should therefore be interpreted as partly reflecting generation quality under this prompt, not only visual grounding.

E. Model Conditioning

Using the same prompt within each dataset condition facilitates standardisation, but it also means the measured performance is a property of the model–prompt pair. The unusually low LLaVA-Med COCO AOPC suggests that the fixed prompt did not reliably induce the intended image-dependent behaviour for that model. The results should therefore be read as one prompting protocol, not as the best grounding each model can achieve; future work should compare model-specific conditioning strategies against a shared-prompt baseline.

F. Data Leakage and Dataset Artifacts

A further limitation is the possibility that MedGemma or LLaVA-Med may have seen some ROCov2 images, or visually similar images, during training. If a model has memorised an image-caption pair, it may reproduce a plausible caption from memory rather than from the visual evidence in the current input. This would reduce the measured probability drops, because even after substantial image perturbation the model could rely on memorised associations or strong language priors instead of the remaining image content. ROCov2 also contains dataset artifacts that complicate the interpretation of saliency maps. Many radiology images include visible

annotations, labels, arrows, or text markers. In such cases, an attribution method may correctly identify a visible label as important evidence for a generated word. For example, if an image contains a label indicating “liver”, the saliency map may highlight that text. After the labelled region is masked, however, the model may still infer the same word from the surrounding anatomy or from the caption context, so the probability drop may remain small. This does not necessarily mean that the saliency method failed; rather, the deletion metric may underestimate the importance of shortcut-like evidence when redundant visual or linguistic cues remain available. This issue is related to prior findings that medical image models can exploit shortcuts or non-pathological cues instead of clinically meaningful image features [20]. In the present setting, such artifacts are especially important because the goal is not only to test whether a model is sensitive to image perturbations, but whether that sensitivity corresponds to clinically meaningful visual grounding. Consequently, saliency maps that highlight visible labels or other annotations should be interpreted with caution, and future work should evaluate the same methods on images with annotation artifacts removed or explicitly stratify examples by the presence of visible text.

VI. CONCLUSION

This thesis empirically benchmarked saliency methods for medical vision-language models. Attention Rollout, GMAR, and Grad-CAM were adapted from ViT classifier settings to VLM architectures to generate per-token saliency maps and were evaluated on two medical VLMs: MedGemma and LLaVA-Med using a teacher-forced deletion test on ROCov2, a radiology dataset. The results show that these methods achieved larger probability drops than a random baseline, indicating that they identify image regions that contribute to the models’ generated tokens. Attention Rollout and GMAR produced nearly identical results, suggesting that weighting attention heads by their gradient is not effective in this VLM setting. Grad-CAM underperformed in comparison to the attention-rollout-based methods, further corroborating the weak gradient signals in VLMs. The ablation experiment demonstrated that visual grounding is affected by model training, prompt strategies, and dataset domain, as probability drops were greater on a general-domain dataset. Overall, the models demonstrated statistically significant visual grounding which was limited by strong language priors. Future work should evaluate model-specific prompts, alternative Grad-CAM hook locations, free-form generation protocols, datasets with fewer visible annotation artifacts, and insertion tests to obtain a more complete picture of visual grounding in medical VLMs.

REFERENCES

- [1] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021. arXiv:2010.11929.
- [2] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626. arXiv:1610.02391.
- [3] S. Abnar and W. Zuidema, “Quantifying attention flow in transformers,” in *Proc. 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4190–4197. arXiv:2005.00928.
- [4] S. Jo, G. Jang, and H. Park, “GMAR: Gradient-driven multi-head attention rollout for vision transformer interpretability,” arXiv preprint arXiv:2504.19414, 2025.
- [5] A. Sellergren *et al.*, “MedGemma technical report,” arXiv preprint arXiv:2507.05201, 2025.
- [6] Gemma Team, Google DeepMind, “Gemma 3 technical report,” arXiv preprint arXiv:2503.19786, 2025.
- [7] J. Rückert *et al.*, “ROCov2: Radiology objects in COntext version 2, an updated multimodal image dataset,” *Scientific Data*, vol. 11, no. 1, p. 688, 2024. arXiv:2405.10004.
- [8] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, and J. Gao, “LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day,” in *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023. arXiv:2306.00890.
- [9] A. Q. Jiang *et al.*, “Mistral 7B,” arXiv preprint arXiv:2310.06825, 2023.
- [10] M. Tschannen *et al.*, “SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features,” arXiv preprint arXiv:2502.14786, 2025.
- [11] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 8748–8763. arXiv:2103.00020.
- [12] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, “Microsoft COCO: Common objects in context,” in *Proc. European Conf. Computer Vision (ECCV)*, 2014, pp. 740–755. arXiv:1405.0312.
- [13] H. Liu, C. Li, Y. Li, and Y. J. Lee, “Improved baselines with visual instruction tuning,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024. arXiv:2310.03744.
- [14] Clinical-AI-Apollo, “Medical-NER: A DeBERTa-v3 token classifier fine-tuned on PubMed for 41 medical entity types,” Hugging Face model repository, 2024. [Online]. Available: <https://huggingface.co/Clinical-AI-Apollo/Medical-NER>
- [15] P. He, J. Gao, and W. Chen, “DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2023. arXiv:2111.09543.
- [16] W. Samek, A. Binder, G. Montavon, S. Lapuschkin, and K.-R. Müller, “Evaluating the visualization of what a deep neural network has learned,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 11, pp. 2660–2673, 2017.
- [17] R. J. Williams and D. Zipser, “A learning algorithm for continually running fully recurrent neural networks,” *Neural Computation*, vol. 1, no. 2, pp. 270–280, 1989.
- [18] A. Saporta, X. Gui, A. Agrawal, A. Pareek, S. Q. H. Truong, C. D. T. Nguyen, V.-D. Ngo, J. Seekins, F. G. Blankenberg, A. Y. Ng, M. P. Lungren, and P. Rajpurkar, “Benchmarking saliency methods for chest X-ray interpretation,” *Nature Machine Intelligence*, vol. 4, no. 10, pp. 867–878, 2022.
- [19] N. Arun, N. Gaw, P. Singh, K. Chang, M. Aggarwal, B. Chen, K. Hoebel, S. Patel, M. Gidwani, J. Adebayo, M. D. Li, and J. Kalpathy-Cramer, “Assessing the trustworthiness of saliency maps for localizing abnormalities in medical imaging,” *Radiology: Artificial Intelligence*, vol. 3, no. 6, p. e200267, 2021.
- [20] A. J. DeGrave, J. D. Janizek, and S.-I. Lee, “AI for radiographic COVID-19 detection selects shortcuts over signal,” *Nature Machine Intelligence*, vol. 3, no. 7, pp. 610–619, 2021.
- [21] L. Barekatain and B. Glocker, “Evaluating the Explainability of Vision Transformers in Medical Imaging,” in *Workshop on Interpretability of Machine Intelligence in Medical Image Computing (iMIMIC), MICCAI 2025*, 2025. arXiv:2510.12021.